

Phase 2 Insight Report

Key insights and findings from E3, M4 and H4

Round 1, Phase 2: Analytical Core

Competition: Data Vortex | AARUUSH'26
Theme: Rebuilding the Social Engine
Team: Entropy
Members: Saanvi Grover & Aditya Sharma
Questions: E3 · M4 · H4
Database: SQLite (Entropy_social_engine.db)

Executive Summary

Each question has a clear literal winner, and each query was built with a benchmark to test whether that winner reflects a real effect.

Question	Answer	How much it can be relied on
E3	Instagram, 4,040.0 average total engagement	All platforms sit within 2.2% of each other. The data could detect a 3.7% gap and finds none ($p = 0.56$).
M4	Instagram, 4,145.2 for the 594 users with 30k+ followers	4.4% above smaller accounts on Instagram, but not significant after correcting for five tests ($p = 0.12$), and the platform ranking reverses between groups.
H4	16 users with under 5k followers in the top 10%	Close to the 13 expected if followers do not matter ($p = 0.38$). 14 are there by posting volume; 2 are genuine per-post outperformers.

HEADLINE

In this dataset, where a post appears and how many followers its author has do not move engagement by more than a few percent. What separates top users is how much they post. That makes two users stand out: `user_ogtvuuki` (Cairo) and `user_kbdvf8d6` (Tokyo), small accounts whose posts rank in the top 10% on quality as well as volume.

E3 – Does Platform Choice Matter?

Rank	Platform	Avg likes	Avg shares	Avg comments	Avg total	vs overall
1	Instagram	2,500.9	1,040.8	499.8	4,040.0	+0.78%
2	YouTube	2,517.8	1,011.8	504.4	4,031.8	+0.58%
3	Facebook	2,528.9	984.2	506.9	4,016.0	+0.18%
4	Reddit	2,488.1	1,002.2	511.2	4,003.9	−0.12%
5	Twitter	2,437.7	1,005.4	506.1	3,951.9	−1.42%

Table 1: E3 result. Overall mean 4,008.7; about 2,000 posts per platform.

Findings

- **Instagram leads on total engagement** (4,040.0), mainly because it has the highest average shares (1,040.8, 5.7% above Facebook’s 984.2). Facebook actually has the most likes (2,528.9).
- **The whole field spans 88 points** (2.2% of the mean), from Instagram to Twitter.
- **This is a measured null, not a small-sample shrug.** With about 1,730 posts per platform, a gap of 150 (3.7%) would be detected 80% of the time. None of the gaps comes close ($p = 0.56$).

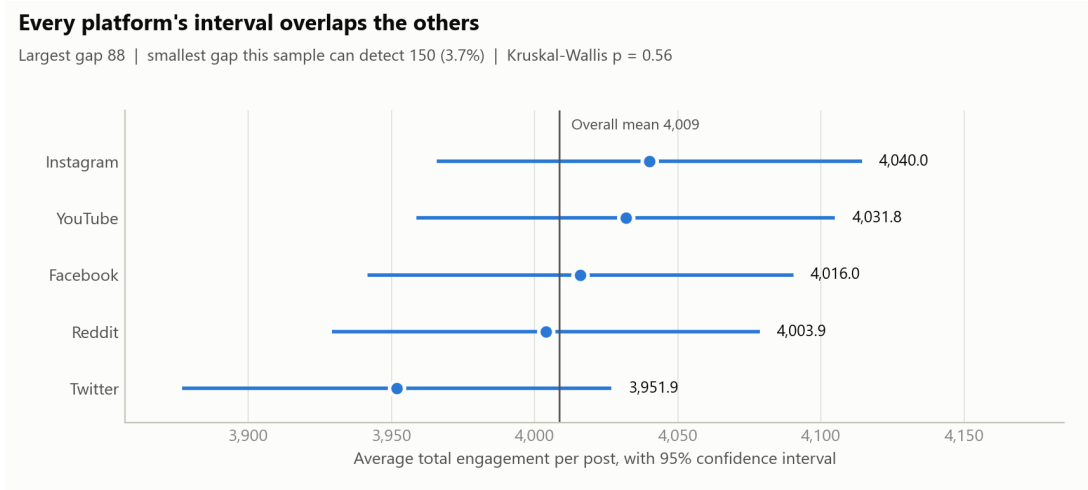


Figure 1: Average total engagement per platform with 95% confidence intervals.

KEY INSIGHT

Platform choice changes expected engagement by less than 4%. Platform decisions should be made on audience fit, cost and content format rather than on expected engagement, because the engagement difference between platforms is too small to justify reallocating content.

M4 – Do Large Accounts Have a Best Platform?

Rank 30k+	Rank <30k	Platform	Avg 30k+	Avg <30k	Lift	Adj. p
1	4	Instagram	4,145.2	3,969.7	+4.42%	0.12
2	2	Facebook	4,000.9	4,025.4	-0.61%	1.00
3	3	Reddit	3,981.4	4,018.3	-0.92%	1.00
4	5	Twitter	3,978.5	3,935.2	+1.10%	1.00
5	1	YouTube	3,947.4	4,090.4	-3.50%	0.25

Table 2: M4 result. 594 users (39.6%) have 30,000+ followers; they account for 3,408 posts with valid engagement.

Findings

- **Large accounts do best on Instagram** (4,145.2 per post), 4.4% more than smaller accounts get there. On its own that difference has $p = 0.023$.
- **It does not hold up.** Testing five platforms makes a p below 0.05 likely somewhere by chance; after Holm correction Instagram's p is 0.12.
- **The ranking reverses.** YouTube is the best platform for smaller accounts and the worst for large ones; Instagram goes from 4th to 1st. The two rankings are negatively related (Spearman $\rho = -0.30$).
- **Across all platforms combined, large and small accounts get the same engagement per post** ($p = 0.98$): lifts on some platforms are cancelled by drops on others.

Instagram's +4.4% for large accounts does not survive correction, and the ranking reshuffles

Mann-Whitney per platform, Holm-adjusted | 30k+ vs under 30k across all platforms $p = 0.98$ | rank agreement $\rho = -0.30$

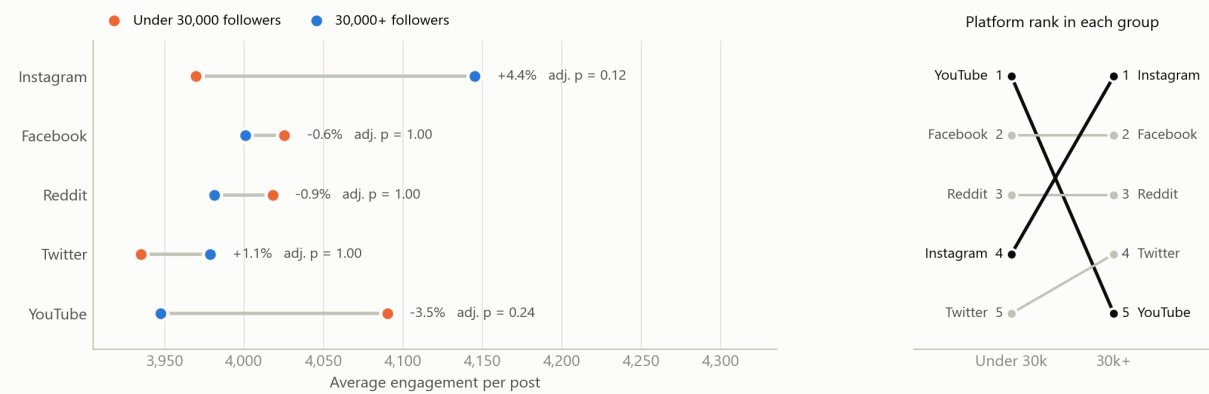


Figure 2: Left: average engagement per post for each group, with lift and Holm-adjusted p . Right: each platform's rank in the two groups.

KEY INSIGHT

A “best platform for influencers” recommendation from this data would not replicate: the ranking that looks best for large accounts is close to the reverse of the one for smaller accounts, and no single difference survives correction. Segment-specific platform strategies need a lift larger than 5% that also holds across time periods before they are worth acting on.

H4 – Are Small Accounts in the Top 10% Anomalous?

#	User	Location	Followers	Posts	Per post	Classification
1	user_uerv85na	Rome, Italy	1,824	16	4,596.5	Volume-driven
2	user_hdas0iau	Rio de Janeiro, Brazil	1,620	16	3,959.1	Volume-driven
3	user_fgjkkrie	Lyon, France	2,211	14	4,477.9	Volume-driven
4	user_n0ok02rt	Dubai, UAE	2,531	18	3,382.0	Volume-driven
7	user_ogtvuuki	Cairo, Egypt	4,459	11	4,987.7	Exceptional per post
16	user_kbdvf8d6	Tokyo, Japan	4,645	10	4,849.0	Exceptional per post

Table 3: Excerpt of the H4 result (top four by total, plus both exceptional users). The full 16-row output is in the SQL Queries PDF and the output screenshot.

Findings

- **16 of the 150 top-decile users have fewer than 5,000 followers.** The threshold for the top 10% is 48,159 total engagement; user_uerv85na (Rome, 1,824 followers) leads with 73,544.
- **That is about what chance predicts.** 8.7% of users have under 5,000 followers, so 13.1 would land in the top decile if followers had no effect. Observing 16 is well within chance ($p = 0.38$). Followers and total engagement are unrelated (Spearman $\rho = -0.01$).
- **Volume explains the list.** The 16 average 12.9 posts against 8.0 for all users, and post count predicts total engagement almost perfectly (Spearman $\rho = 0.91$). 14 of the 16 have ordinary per-post engagement.
- **Two genuine outperformers.** user_ogtvuuki (Cairo; 4,987.7 per post) and user_kbdvf8d6 (Tokyo; 4,849.0 per post) are in the top 10% on both total and per-post engagement with under

5,000 followers.

- **Small accounts are not over-represented among per-post standouts.** 12.5% of the 16 are top 10% per post, compared with 15.3% of the whole top decile.

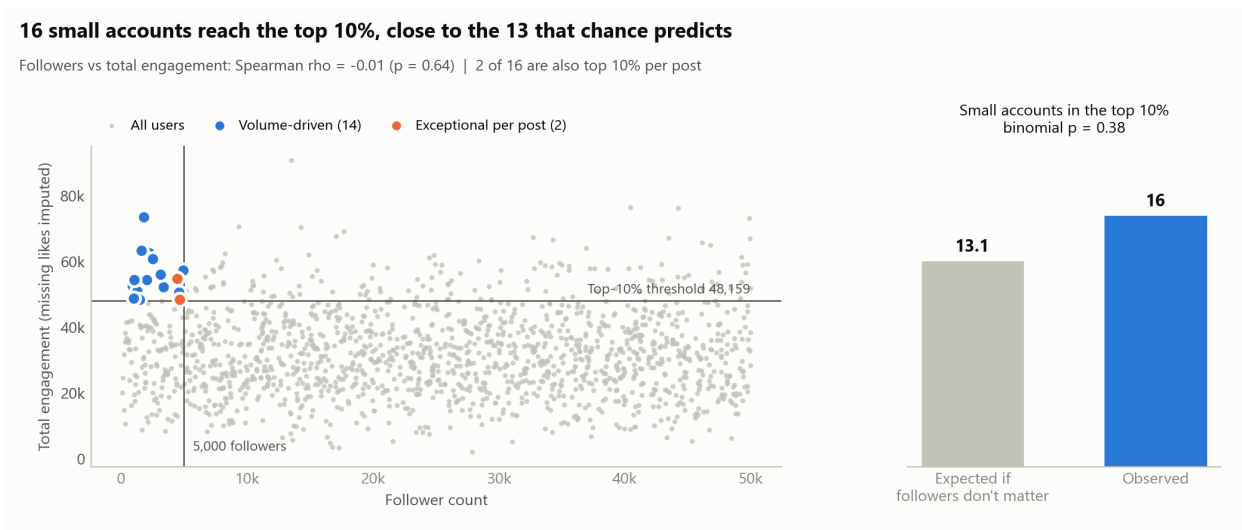


Figure 3: Left: followers against total engagement; small-account anomalies highlighted. Right: observed count against the count expected if followers do not matter.

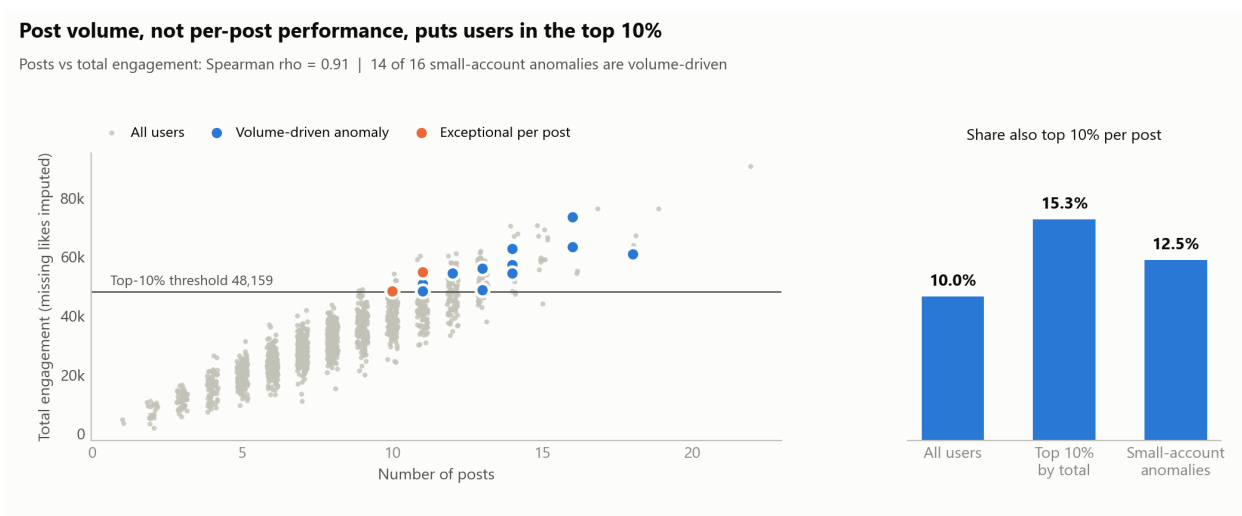


Figure 4: Left: total engagement rises almost linearly with post count. Right: share of each group that is also top 10% per post.

KEY INSIGHT

A top-10% list built on totals is mostly a list of frequent posters. Screening for “high-performing or suspicious” small accounts should rank on per-post engagement with a minimum post count, which narrows 16 names to 2. Those two are worth a manual review; the other 14 are not anomalous.

Cross-Question Synthesis

Lever	Largest effect	Evidence	Verdict
Platform (E3)	2.2% spread	$p = 0.56$; detectable gap 3.7%	Below 4%
Audience size by platform (M4)	+4.4% (Instagram)	Holm $p = 0.12$; ranking reverses	Not stable
Audience size on totals (H4)	$\rho = -0.01$	$p = 0.64$; 16 vs 13.1 expected	No effect
Posting volume on totals (H4)	$\rho = 0.91$	Strongest relationship found	Dominant driver

These results agree with the Phase 1 EDA, where 5 of 6 hypothesis tests found no effect and most dimensions varied by under 10%. Phase 2 adds three things: an upper bound on the platform effect (under 4%), evidence that a nominally significant segment result disappears under correction, and a user-level driver that does matter (posting volume).

Recommendations

1. **Treat platform differences under 4% as noise.** With samples of this size, report platform comparisons with confidence intervals and do not reallocate content on gaps below the detectable threshold.
2. **Correct for multiple comparisons before acting on segment results.** Testing every platform for a segment will usually produce one “significant” result by chance, as M4’s Instagram result shows.
3. **Rank users on per-post engagement with a minimum post count.** Totals mostly measure activity. Applied to H4, this reduces 16 flagged accounts to 2 worth reviewing: `user_ogtvuuki` and `user_kbdvf8d6`.
4. **Impute corrupted likes for user-level metrics instead of zero-filling.** Zero-filling changed 2 of the 16 names in H4.
5. **Investigate the multi-platform effect next.** It remains the only lever with a significant effect in Phase 1 (about 15% higher engagement) and none of these three questions tests it.

Limitations

- The dataset appears to be synthetic (Phase 1 found near-uniform distributions and template-generated text), so effect sizes may not transfer to real platforms.
- Imputation assumes corrupted likes are missing at random, as Phase 1 found; if corruption targeted particular users, per-user averages would be biased.
- H4 classifications depend on the 10% cut-off; users just outside the decile boundaries could move with small data changes.